Google Agent Development Kit

Build, run, evaluate, and deploy production agents

GEN AI GURU

Independent educational explainer

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ADK is an open-source production agent framework

Build production agents, not prototypes, across five official language SDKs.

- Python, TypeScript, Go, Java, and Kotlin.
- Start with a prompt and tools, then grow into graphs and multi-agent workflows.
- Context is assembled like source code, not pasted until overflow.

Official ADK sources: [1] [2]

Eight primitives define the runtime

- Agent: LlmAgent for reasoning, or workflow agents for deterministic control.
- Tool, Callback, Session, State, Memory, Artifact, Event, Runner.

Takeaway Session State is this conversation. Memory is the long-term archive.

Official ADK sources: [2] [6] [7]

Python path: install, scaffold, run

- Python 3.10+ and `pip install google-adk`.
- `adk create` writes `agent.py` with a required `root_agent`.
- `adk run` for CLI. `adk web` for a local chat UI.
- ADK Web is documented as development-only, not production.

Official ADK sources: [3]

Tools execute. Models reason.

- Function tools wrap developer code. Docstrings drive selection.
- MCP, OpenAPI, and AgentTool extend the surface.
- Confirm dangerous actions. Keep long-running work asynchronous.
- Skills load instructions on demand; the feature is experimental.

Official ADK sources: [5] [17]

Workflows and ADK 2.0 graphs

- Sequential, Parallel, and Loop agents orchestrate without asking a model to route.
- Graphs add nodes, edges, and explicit routing.
- Python ADK 2.0 GA: 19 May 2026. Review breaking changes before upgrading 1.x.

Official ADK sources: [4] [11] [12] [14]

Deploy, evaluate, then govern

- Agent Runtime, Cloud Run, GKE, or any container host.
- Score both the final answer and the tool trajectory.
- Identity, guardrails, sandboxes, VPC-SC, and A2A for remote contracts.

Takeaway Pilot one LlmAgent and one tool. Expand from measured trajectories.

Official ADK sources: [9] [10] [13] [16]

Trusted agents are designed, evaluated, and governed.

Pilot one agent. Measure the path. Then add graphs, memory, and deployment.